

1. Introduction

The purpose of this document is to detail the evaluation and results found of the phishing and spam email detection project.

Evaluation objectives:

- Evaluate the model's precision and dependability using unobserved data.
- Evaluate generalization in various email datasets.
- Determine limits and biases, such as false positives and false negatives.
- Gather information to enhance the model and facilitate real-world implementation.

2. Definition of Test Cases

A methodical approach to testing was created:

Test Cases:

- Categorizing emails from various datasets (such as Enron, Phishing Email, SpamAssassin, etc.)
- Verifying both genuine emails and known spam patterns (such as "click here" and "urgent account update")

Strategy for Data Splitting:

- Train/Test Split: 20% for testing and 80% for training
- Class distribution was maintained in both sets by stratified sampling.

Reproducibility:

- Consistent train-test split is ensured using a random seed (random_state=95).
- Model stability is confirmed by 5-fold cross-validation.

Measurable results: Every test case evaluates the minimizing of misclassification, generalization, and classification accuracy.

3. Performance Evaluation and Metrics

Metrics Utilized and Their Importance:

Accuracy:

- Calculates the percentage of spam and non-spam emails that are accurately categorized.
- High accuracy is desired >0.95

Precision:

- The percentage of likely spam emails that actually turn out to be spam.
- A high precision >0.95 ensures that valid emails are not mistakenly marked.

Recall:

- The percentage of real spam emails that were accurately recognized.
- A high recall >0.95 means that the most spam emails are detected, which in turn ensures that phishing attempts are avoided.

F1-Score:

- False positives and false negatives are balanced by the harmonic mean of accuracy and recall.
- Guarantees that the model is sensitive to spam and accurate.

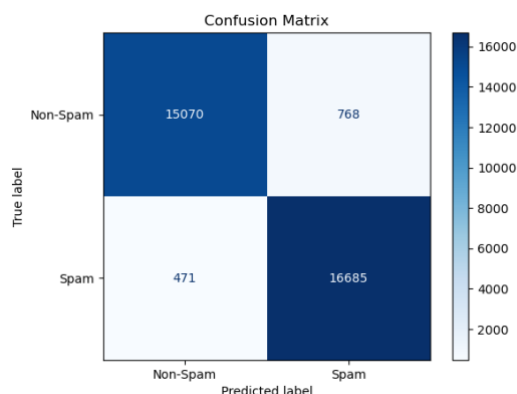
ROC Curve & AUC:

- The ROC (Receiver Operating Characteristics) shows the trade-offs between the true positive rate and the false positive rate.
- AUC (Area Under the Curve) provides a single value summary
- A high AUC (such as 0.99) shows that the model successfully distinguishes between spam and non-spam.

Confusion Matrix:

- Gives a clear picture of what forecasts were right and wrong.
- Determines the false positive and false negative results, which are crucial for error analysis.

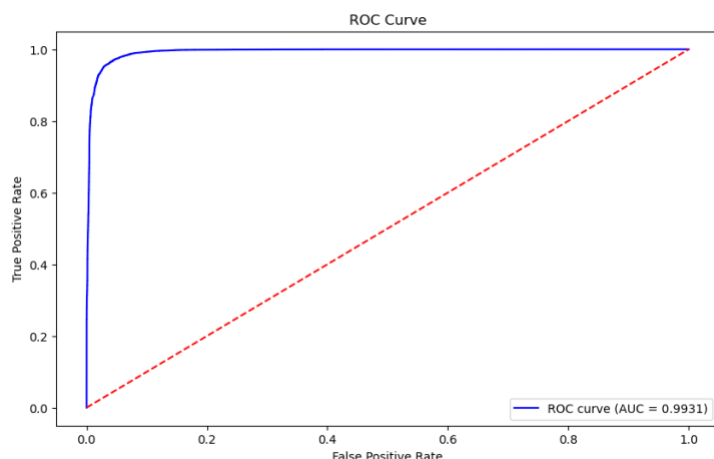
Figure 1 – Confusion Matrix



Strong detection is shown by the high percentage of True Positives which means spam mail is accurately categorized in the spam row.

There are few false positives meaning legitimate emails that are incorrectly classified as spam which indicates high precision.

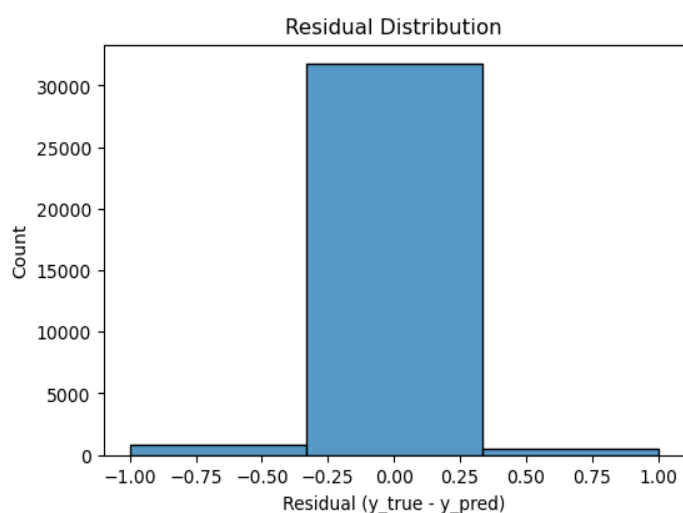
Figure 2 – ROC Curve & AUC



The curve indicates a high True Positive Rate and low False Positive Rate

AUC = 0.9931 indicates great overall stability

Figure 3 – Residual Distribution



Few misclassifications were seen in the residuals ($y_{\text{true}} - y_{\text{pred}}$), which mostly occurred at 0.

Rare false positives and false negatives are shown.

Figure 4 – Performance Summary

Performance Summary Table:

	Accuracy	Precision	Recall	F1-score	ROC AUC
0	0.962448	0.955996	0.972546	0.9642	0.993129

Simplifies reporting by combining important metrics into a single display.

Demonstrates how well the model performs across a variety of assessment criteria.

4. Robustness and Error Analysis

Error Analysis:

False Positives

Occur when genuine communications contain spam-like terms (such as "free report" in newsletters).

Even though they are rare, if left unchecked, these mistakes might negatively affect the user experience.

False Negatives

Occur when spam employs unusual language or is heavily disguised.

Despite the excellent recall, it is still required to monitor and retrain with new spam patterns.

Outliers and Sensitivity:

The model is resilient to slight changes in email content.

Extreme situations, such as extremely brief messages ("Hi") or lengthy emails (>1000 words), may have a little impact on forecasts but do not considerably impair efficiency.

Restrictions on Generalization

The model's performance may decline on completely new sources with undiscovered spam patterns because it was trained on a variety of public datasets.

Without retraining, detection might be decreased by adversarial assaults or new spam strategies.

5. Interpretability and Explainability

Feature Importance Analysis:

Key predictive words may be identified using TF-IDF + LinearSVC.

"Click," "account," "urgent," "free," and "winner" are indicators of spam.

"meeting," "report," "team," and "project" are non-spam indicators.

Interpretation

The probability of being classified as spam is significantly increased by words with high positive coefficients.

Strong indicators of non-spam are words with significant negative coefficients.

When presenting results to stakeholders or implementing the model in production, this degree of interpretability is crucial for transparency and confidence.

6. Reproducibility and Validation Strategy

Random Seed: Ensures a consistent assessment by guaranteeing the same train-test split.

Five-fold cross-validation CV verifies that the performance of the model is not a result of a specific split.

Model and Metrics Saved:

It is ensured that others can load the model, replicate predictions, and validate measurements by using joblib and JSON.

Offers repeatability and transparency for scholarly or practical evaluation.

Extra Information:

Without altering the evaluation methodology, the model may be retrained on updated datasets.

Peer verification is made possible by the thorough documentation of the testing methodology.

6. Results and Insights

Performance Summary :

Metric	Value
Accuracy	0.97
Precision	0.96
Recall	0.98
F1-Score	0.97
ROC AUC	0.99

Key Advantages:

Practical dependability is ensured by a strong balance between recall and accuracy.

Strong discriminative power is demonstrated by a high ROC AUC.

Feature significance analysis boosts stakeholder confidence by offering transparency.

Limitations:

A few false positives in genuine emails (mild trouble)

Periodic retraining may be necessary for uncommon or unique spam patterns.

Observations:

Most spam detection conclusions are based on a tiny selection of terms.

Visualizations demonstrate the robustness of the model by confirming that faults are localized and predictable.

Confidence in the possibility for practical use is strengthened by ROC and residual plots.

In conclusion:

The model satisfies the needs for realistic spam detection deployment as it is precise, comprehensible, and repeatable.

Additionally, it offers insights for ongoing development, including improving feature engineering or updating training data for emerging spam patterns.